IT 3010 – Network Design and Management

Practical 02 – Lab Report

IT23565494 – S.J.M.M.R. Samarakoon

1. Installing DHCP server in server machine :

```
root@mlb-dc1-centos7:~
File  Edit  View  Search  Terminal  Help
  Updating   : 12:dhcp-common-4.2.5-83.el7.centos.2.x86_64                2/7
  Installing : 12:dhcp-4.2.5-83.el7.centos.2.x86_64                       3/7
  Updating   : 12:dhclient-4.2.5-83.el7.centos.2.x86_64                   4/7
  Cleanup    : 12:dhclient-4.2.5-82.el7.centos.x86_64                     5/7
  Cleanup    : 12:dhcp-common-4.2.5-82.el7.centos.x86_64                  6/7
  Cleanup    : 12:dhcp-libs-4.2.5-82.el7.centos.x86_64                    7/7
  Verifying  : 12:dhcp-4.2.5-83.el7.centos.2.x86_64                       1/7
  Verifying  : 12:dhclient-4.2.5-83.el7.centos.2.x86_64                   2/7
  Verifying  : 12:dhcp-libs-4.2.5-83.el7.centos.2.x86_64                  3/7
  Verifying  : 12:dhcp-common-4.2.5-83.el7.centos.2.x86_64                4/7
  Verifying  : 12:dhcp-common-4.2.5-82.el7.centos.x86_64                  5/7
  Verifying  : 12:dhclient-4.2.5-82.el7.centos.x86_64                     6/7
  Verifying  : 12:dhcp-libs-4.2.5-82.el7.centos.x86_64                    7/7

Installed:
  dhcp.x86_64 12:4.2.5-83.el7.centos.2

Dependency Updated:
  dhclient.x86_64 12:4.2.5-83.el7.centos.2
  dhcp-common.x86_64 12:4.2.5-83.el7.centos.2
  dhcp-libs.x86_64 12:4.2.5-83.el7.centos.2

Complete!
[root@mlb-dc1-centos7 ~]#
```

2. Configuring DHCP settings :

```
File  Edit  View  Search  Terminal  Help
[svr1@mlb-dc1-centos7 ~]$ vi /etc/sysconfig/dhcpd
[svr1@mlb-dc1-centos7 ~]$ vi /etc/dhcp/dhcpd.conf
[svr1@mlb-dc1-centos7 ~]$ su -
Password:
Last login: Sat Feb  7 12:33:39 EST 2026 on pts/0
[root@mlb-dc1-centos7 ~]# vi /etc/dhcp/dhcpd.conf
[root@mlb-dc1-centos7 ~]# vi /etc/dhcp/dhcpd.conf
[root@mlb-dc1-centos7 ~]#
```

```
File  Edit  View  Search  Terminal  Help

# DHCP Server Configuration file.
#   see /usr/share/doc/dhcp*/dhcpd.conf.example
#Internal Subnet Configuration
subnet 10.0.1.0 netmask 255.255.255.0 {
  range 10.0.1.20 10.0.1.30;
  option domain-name-servers 10.0.1.2;
  option domain-name "dsnm.sub";
  option routers 10.0.1.1;
  option broadcast-address 10.0.1.255;
  default-lease-time 600;
  max-lease-time 7200;
}
~
~
~
~
~
~
~
```

3. Starting DHCP server :

```
File  Edit  View  Search  Terminal  Help
[svr1@mlb-dc1-centos7 ~]$ vi /etc/sysconfig/dhcpd
[svr1@mlb-dc1-centos7 ~]$ vi /etc/dhcp/dhcpd.conf
[svr1@mlb-dc1-centos7 ~]$ su -
Password:
Last login: Sat Feb  7 12:33:39 EST 2026 on pts/0
[root@mlb-dc1-centos7 ~]# vi /etc/dhcp/dhcpd.conf
[root@mlb-dc1-centos7 ~]# vi /etc/dhcp/dhcpd.conf
[root@mlb-dc1-centos7 ~]# exit
logout
[svr1@mlb-dc1-centos7 ~]$ systemctl start dhcpd
```

4. Checking status of DHCP server :

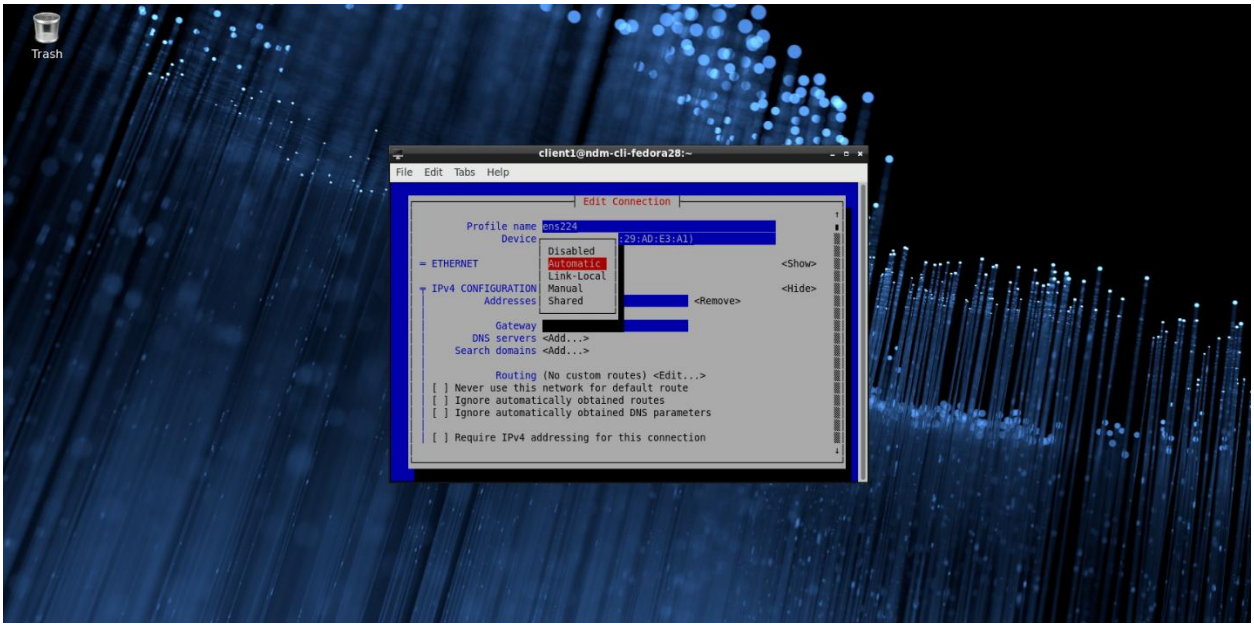
```
[root@mlb-dc1-centos7 ~]# systemctl status dhcpd
● dhcpd.service - DHCPv4 Server Daemon
   Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; disabled; vendor preset: disabled)
   Active: active (running) since Tue 2026-02-10 22:02:38 EST; 1min 39s ago
     Docs: man:dhcpd(8)
           man:dhcpd.conf(5)
 Main PID: 4262 (dhcpd)
   Status: "Dispatching packets..."
    Tasks: 1
   CGroup: /system.slice/dhcpd.service
           └─4262 /usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid

Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk dhcpd[4262]: Sending on   LPF/ens34/00:0c:29:00:d2:f7/10.0.1.0/24
Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk dhcpd[4262]:
Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk dhcpd[4262]: No subnet declaration for ens33 (192.168.112.128).
Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk dhcpd[4262]: ** Ignoring requests on ens33.  If this is not what
Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk dhcpd[4262]:    you want, please write a subnet declaration
Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk dhcpd[4262]:    in your dhcpd.conf file for the network segment
Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk dhcpd[4262]:    to which interface ens33 is attached. **
Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk dhcpd[4262]:
Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk dhcpd[4262]: Sending on   Socket/fallback/fallback-net
Feb 10 22:02:38 mlb-dc1-centos7.ndm.lk systemd[1]: Started DHCPv4 Server Daemon.
[root@mlb-dc1-centos7 ~]#
```

5. Enabling DHCP on boot :

```
[root@mlb-dc1-centos7 ~]# systemctl enable dhcpd
Created symlink from /etc/systemd/system/multi-user.target.wants/dhcpd.service to /usr/lib/systemd/system/dhcpd.service.
[root@mlb-dc1-centos7 ~]#
```

6. Changing client IP settings to automatic :



7. Assigning a temporary IP address to ens224:

```
[client1@ndm-cli-fedora28 ~]$ su -
Password:
Last login: Wed Feb  4 08:29:19 +0530 2026 on pts/0
[root@ndm-cli-fedora28 ~]# ifconfig ens224 10.0.1.99 netmask 255.255.255.0 up
[root@ndm-cli-fedora28 ~]#
```

```
[root@ndm-cli-fedora28 ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.112.130  netmask 255.255.255.0  broadcast 192.168.112.255
        inet6 fe80::3d7f:9df9:f850:746f  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:ad:e3:97  txqueuelen 1000  (Ethernet)
        RX packets 363  bytes 33918 (33.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 129  bytes 12695 (12.3 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ens224: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.1.99  netmask 255.255.255.0  broadcast 10.0.1.255
        inet6 fe80::6b55:933a:f6e5:af9c  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:ad:e3:a1  txqueuelen 1000  (Ethernet)
        RX packets 36  bytes 4678 (4.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 136  bytes 24348 (23.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[root@ndm-cli-fedora28 ~]#
```

8. Rebooting after reassign of temporary IP :

client1@ndm-cli-fedora28:~

File Edit Tabs Help

```
[client1@ndm-cli-fedora28 ~]$ ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.112.130  netmask 255.255.255.0  broadcast 192.168.112.255
        inet6 fe80::3d7f:9df9:f850:746f  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:ad:e3:97  txqueuelen 1000  (Ethernet)
        RX packets 26  bytes 2598 (2.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 50  bytes 6221 (6.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ens224: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.1.20  netmask 255.255.255.0  broadcast 10.0.1.255
        inet6 fe80::6b55:933a:f6e5:af9c  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:ad:e3:a1  txqueuelen 1000  (Ethernet)
        RX packets 1  bytes 346 (346.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 32  bytes 4663 (4.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 4  bytes 240 (240.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4  bytes 240 (240.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[client1@ndm-cli-fedora28 ~]$
```